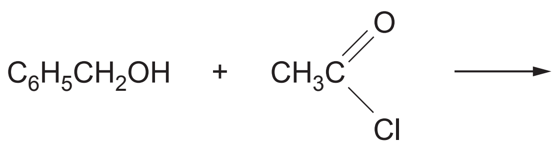
Examiner only

SECTION A

Answer **all** questions.

1. Phenylmethyl ethanoate, used in perfumery, is made by reacting phenylmethanol and ethanoyl chloride.

Complete the equation for this reaction, showing the structure of the organic product. [1]



2. An azo dye has a maximum absorption of visible light at 403 nm.

(a) Calculate the frequency corresponding to this wavelength. [1]

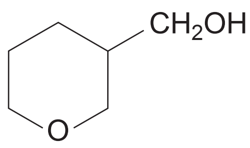
Frequency = ... Hz

(b) Explain why this dye is yellow in white light. [1]

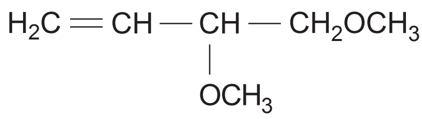
....

....

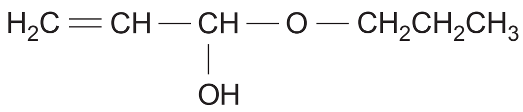
3. Four isomers of formula $C_6H_{12}O_2$ are given below.



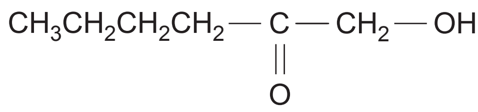
compound **D**



compound **E**



compound **F**



compound **G**

Deduce which isomer has an infrared absorption spectrum showing absorptions at 1620 cm^{-1} and at 3200–3500 cm^{-1} but **not** at 1650–1750 cm^{-1}. Give reasons for your answer. [2]

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4. Give the formula of the triphenylmethyl radical. [1]

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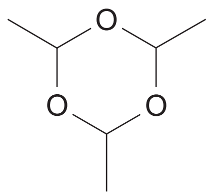
5. The structure of paraldehyde is shown below.



Explain what would be seen in the ^{13}C NMR spectrum of this compound.

References to the positions of the chemical shifts are **not** required. [2]

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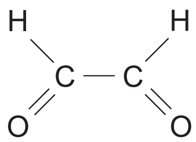
6. Ethanal is oxidised by selenium dioxide giving ethanedial.

CH_3CHO + SeO_2 ⟶  + Se + H_2O

Calculate the atom economy of this reaction to produce ethanedial.

Give your answer to an **appropriate** number of significant figures. [2]

Atom economy = ... %

10

